SCS CVX Julia DCP - disciplined convex programming SOCP
Norm approximation

1 Norm Approximation

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$. The objective is to minimize the residual vector $\mathbf{r} = \mathbf{A}\mathbf{x} - \mathbf{b}$.

The general problem is formulated as,

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|.$$

Definition 1.1 (ℓ_1 norm): *For $\mathbf{y} \in \mathbb{R}^m$, ℓ_1 norm is defined as*

$$\|\mathbf{y}\|_1 = \sum_{i=1}^{m} |y_i|$$

Definition 1.2 (ℓ_2 norm): *For $\mathbf{y} \in \mathbb{R}^m$, ℓ_2 norm is defined as*

$$\|\mathbf{y}\|_2 = \left(\sum_{i=1}^{m} y_i^2 \right)^{1/2}$$

Definition 1.3 (ℓ_∞ norm): *For $\mathbf{y} \in \mathbb{R}^m$, ℓ_∞ norm is defined as*

$$\|\mathbf{y}\|_\infty = \max_{1 \le i \le m} |y_i|$$

1.1 Penalty Functions

1.1.1 Norm Based ($p = 1, 2, \infty$)

$\| \cdot \|_1, \| \cdot \|_2, \| \cdot \|_\infty$.

1.1.2 Dead Zone Linear

Functions of the form, for $r \in \mathbb{R}^n$,

$$\sum_i \phi(r_i),$$

where

$$\phi(r) = \max\{|r| - a, 0\}.$$

1.1.3 Log Barrier

$$\phi(u) = \begin{cases} -\log(1 - u^2), & |u| < 1 \\ \infty, & |u| \ge 1 \end{cases}$$

1.1.4 Huber Penalty

2 Moreau Envelope

Definition 2.1 (Moreau Envelope): *Given $f : \mathbb{R}^n \to \mathbb{R}$, we have the Moreau Envelope as*

$$f_u(\mathbf{x}) = \inf_{\mathbf{y}\in\mathbb{R}^n} \left\{ f(\mathbf{y}) + \frac{1}{2u}\|\mathbf{x}-\mathbf{y}\|^2 \right\}.$$

Remark 2.2: *The above expression is essentially an infimal convolution of $uf : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$ where $g = (1/2)\|\cdot\|_2^2$.*

Definition 2.3 (Proximal Operator): *The proximal operator $prox_u(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}^n$ of f with parameter u is defined as,*

$$prox_u(\mathbf{x}) = \arg\min_{\mathbf{y}\in\mathbb{R}^n} \left\{ f(\mathbf{y}) + \frac{1}{2u}\|\mathbf{x}-\mathbf{y}\|^2 \right\}.$$

2.0.1 Properties of Moreau Envelope

1. $f_u(\mathbf{x})$ is differentiable if f is continuous. Further, the derivative is continuous.
2. $f_u(\mathbf{x})$ is defined for all $\mathbf{x} \in \mathbb{R}^n$, even when f is not.
3. $f_u(\mathbf{x})$ is convex whenever f is convex.
4. If x^* is the unique minimizer of f, then it is the unique minimizer of $f_u(\mathbf{x})$

Remark 2.4: *$f_u(\mathbf{x})$ is convex, since it is the min over one variable of a jointly convex function.*

Remark 2.5: *$f_u(\mathbf{x})$ is continuously smooth.*

Remark 2.6:

$$epi\, f_u(x) = epi\, f(x) \oplus epi \left(\frac{1}{2u}\|x\|^2 \right)$$

Remark 2.7: *x^* is a minima of f if and only if x^* is a minima of f_u.*

Remark 2.8:

$$\nabla f_u(\mathbf{x}) = \frac{1}{u}(1 - prox_u(x)).$$

3 Epigraphs

TODO: complete this
Duality
Supporting Hyperplane Theorem

Definition 3.1 (Conjugate of a function): *Let $f : \mathbb{R}^n \to \mathbb{R}$. The conjugate $f^* : \mathbb{R}^n \to \mathbb{R}$ of f is defined as,*

$$f^*(\mathbf{y}) = \sup_{\mathbf{x}\in dom f} (\mathbf{y}^T\mathbf{x} - f(\mathbf{x})).$$

Remark 3.2: *The conjugate of f is also called Legendre transform, polar of f, Fenchel dual.*

Remark 3.3: *f^* is convex, irrespective of f.*

Remark 3.4:

$$-f^*(0) = \min_{\mathbf{x}\in\mathbb{R}^n} f(\mathbf{x})$$

Remark 3.5: *Fenchel Young inequality*

$$f(\mathbf{x}) + f^*(\mathbf{m}) \geq \mathbf{m}^T\mathbf{x}$$

3.1 Recovering f from f^*

Definition 3.6 (Biconjugate): *Let $f^* : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be the conjugate of a function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$. The **biconjugate** $f^{**} : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is defined as the conjugate of f^*,*

$$f^{**}(\mathbf{x}) = (f^*)^*(\mathbf{x}) = \sup_{\mathbf{y}\in\mathbb{R}^n} \left(\mathbf{y}^T\mathbf{x} - f^*(\mathbf{y})\right).$$

Remark 3.7 (Geometric Interpretation): *The biconjugate f^{**} constructs the pointwise supremum (upper envelope) of all affine minorants of f. Specifically, the epigraph of f^{**} is the **closed convex hull** of the epigraph of f,*

$$epi\, f^{**} = cl(conv(epi\, f)).$$

Thus, f^{**} represents the "tightest" closed convex function that minorizes f.

3.1.1 Properties of the Biconjugate

Corollary 3.8 (Weak Duality): *For any proper function f, the inequality $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$ holds for all $\mathbf{x} \in \mathbb{R}^n$.*

Proof. Recall the **Fenchel-Young inequality**, which states that for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$f^*(\mathbf{y}) + f(\mathbf{x}) \geq \mathbf{x}^T\mathbf{y}.$$

Rearranging terms yields:

$$\mathbf{x}^T\mathbf{y} - f^*(\mathbf{y}) \leq f(\mathbf{x}).$$

Since this inequality holds for all $\mathbf{y} \in \mathbb{R}^n$, it must also hold for the supremum over $\mathbf{y}$:

$$\sup_{\mathbf{y}\in\mathbb{R}^n} \left(\mathbf{x}^T\mathbf{y} - f^*(\mathbf{y})\right) \leq f(\mathbf{x}).$$

By definition of the biconjugate, this is exactly $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$. $\square$

Theorem 3.9 (Fenchel-Moreau): *Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be a proper function. Then $f = f^{**}$ if and only if f is **closed** (lower semi-continuous) and **convex**.*

Proof. The forward implication is immediate: since f^{**} is defined as the pointwise supremum of a family of affine functions, it is necessarily closed and convex. Thus, if $f = f^{**}$, f must share these properties.

For the reverse implication, assume f is proper, closed, and convex. By the weak duality lemma, $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$. It remains to show that $f(\mathbf{x}) \leq f^{**}(\mathbf{x})$.

We proceed by contradiction. Suppose there exists a point $\mathbf{x}_0 \in \mathbb{R}^n$ such that $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0)$. This implies that the point $(\mathbf{x}_0, f^{**}(\mathbf{x}_0)) \in \mathbb{R}^{n+1}$ lies strictly below the epigraph of f, denoted $\mathrm{epi}(f)$. Since f is closed and convex, $\mathrm{epi}(f)$ is a closed convex set.

By the **Strict Separating Hyperplane Theorem**, there exists a non-zero vector $(\mathbf{y}, \mu) \in \mathbb{R}^n \times \mathbb{R}$ and a scalar $\alpha \in \mathbb{R}$ such that:

$$\mathbf{y}^T\mathbf{x}_0 + \mu f^{**}(\mathbf{x}_0) > \alpha > \mathbf{y}^T\mathbf{x} + \mu t, \quad \forall(\mathbf{x}, t) \in \mathrm{epi}(f).$$

Since $(\mathbf{x}, t) \in \mathrm{epi}(f)$ allows t to be arbitrarily large, we must have $\mu \leq 0$ (otherwise the RHS would be unbounded).

Case 1: Non-vertical separation ($\mu < 0$) Divide the inequality by $-\mu > 0$. Let $\mathbf{m} = \mathbf{y}/(-\mu)$ and $c = \alpha/(-\mu)$. The separation becomes:

$$\mathbf{m}^T\mathbf{x}_0 - f^{**}(\mathbf{x}_0) > c > \mathbf{m}^T\mathbf{x} - t, \quad \forall(\mathbf{x}, t) \in \mathrm{epi}(f).$$

Setting $t = f(\mathbf{x})$ for $\mathbf{x} \in \mathrm{dom}(f)$, we obtain $\mathbf{m}^T\mathbf{x} - f(\mathbf{x}) < c$. Taking the supremum over $\mathbf{x}$:

$$f^*(\mathbf{m}) = \sup_{\mathbf{x}}(\mathbf{m}^T\mathbf{x} - f(\mathbf{x})) \leq c.$$

However, the left side of the separation inequality yields $\mathbf{m}^T\mathbf{x}_0 - c > f^{**}(\mathbf{x}_0)$. This implies that the affine function $h(\mathbf{z}) = \mathbf{m}^T\mathbf{z} - f^*(\mathbf{m})$ satisfies:

$$h(\mathbf{x}_0) = \mathbf{m}^T\mathbf{x}_0 - f^*(\mathbf{m}) \geq \mathbf{m}^T\mathbf{x}_0 - c > f^{**}(\mathbf{x}_0).$$

This contradicts the definition of f^{**} as the supremum of *all* such affine minorants: $f^{**}(\mathbf{x}_0) \geq h(\mathbf{x}_0)$.

Case 2: Vertical separation ($\mu = 0$) If $\mu = 0$, the separation reduces to $\mathbf{y}^T\mathbf{x}_0 > \alpha > \mathbf{y}^T\mathbf{x}$ for all $\mathbf{x} \in \mathrm{dom}(f)$. This means $\mathbf{x}_0$ is strictly separated from the domain of f in $\mathbb{R}^n$. Since f is proper and convex, it admits at least one affine minorant $g(\mathbf{z}) = \mathbf{a}^T\mathbf{z} - b \leq f(\mathbf{z})$. Consider the function $g_k(\mathbf{z}) = g(\mathbf{z}) + k(\mathbf{y}^T\mathbf{z} - \alpha)$. For $\mathbf{x} \in \mathrm{dom}(f)$, since $\mathbf{y}^T\mathbf{x} < \alpha$, the term $k(\mathbf{y}^T\mathbf{x} - \alpha)$ is negative. Thus, $g_k(\mathbf{x}) \leq g(\mathbf{x}) \leq f(\mathbf{x})$ for all $k > 0$. This means g_k is an affine minorant of f for any k. At $\mathbf{x}_0$, however, $\mathbf{y}^T\mathbf{x}_0 > \alpha$, so the term $k(\mathbf{y}^T\mathbf{x}_0 - \alpha) \to +\infty$ as $k \to \infty$. Consequently, $f^{**}(\mathbf{x}_0) \geq g_k(\mathbf{x}_0) \to +\infty$. Since we assumed $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0) < \infty$ (if $f(\mathbf{x}_0) = \infty$ the inequality holds trivially), this yields a contradiction.

Thus, the assumption $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0)$ is false. We conclude $f = f^{**}$. $\square$

Remark 3.10: *If f is convex but not closed, f^{**} is the lower semi-continuous closure of f. If f is not convex, f^{**} is the convex envelope of f.*

4 Duality

Corollary 4.1: *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex and differentiable function. For any $\mathbf{x}, \mathbf{m} \in \mathbb{R}^n$, the following statements are equivalent:*

1. $\mathbf{m} = \nabla f(\mathbf{x})$
2. $f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T\mathbf{x}$

Proof. Recall the Fenchel-Young inequality, which states $f(\mathbf{x}) + f^*(\mathbf{m}) \geq \mathbf{m}^T\mathbf{x}$ for all $\mathbf{x}, \mathbf{m}$. We prove the equivalence of the equality condition and the gradient condition.

(1 $\implies$ 2): Assume $\mathbf{m} = \nabla f(\mathbf{x})$. Since f is convex and differentiable, the first-order condition for convexity states that for all $\mathbf{y} \in \mathbb{R}^n$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}).$$

Substituting $\mathbf{m} = \nabla f(\mathbf{x})$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{m}^T\mathbf{y} - \mathbf{m}^T\mathbf{x}.$$

Rearranging terms to isolate $\mathbf{m}^T\mathbf{y} - f(\mathbf{y})$,

$$\mathbf{m}^T\mathbf{y} - f(\mathbf{y}) \leq \mathbf{m}^T\mathbf{x} - f(\mathbf{x}).$$

Taking the supremum of the left-hand side over all $\mathbf{y} \in \mathbb{R}^n$ yields the definition of the conjugate $f^*(\mathbf{m})$,

$$f^*(\mathbf{m}) = \sup_{\mathbf{y}}(\mathbf{m}^T\mathbf{y} - f(\mathbf{y})) \leq \mathbf{m}^T\mathbf{x} - f(\mathbf{x}).$$

This implies $f(\mathbf{x}) + f^*(\mathbf{m}) \leq \mathbf{m}^T\mathbf{x}$. Combining this with the Fenchel-Young inequality (which gives the reverse inequality $\geq$), we obtain equality,

$$f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T\mathbf{x}.$$

(2 $\implies$ 1): Assume $f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T\mathbf{x}$. Rearranging for $f(\mathbf{x})$,

$$f(\mathbf{x}) = \mathbf{m}^T\mathbf{x} - f^*(\mathbf{m}).$$

By definition of the conjugate, $f^*(\mathbf{m}) \geq \mathbf{m}^T\mathbf{y} - f(\mathbf{y})$ for any $\mathbf{y} \in \mathbb{R}^n$. Substituting this inequality into the equation above,

$$f(\mathbf{x}) \leq \mathbf{m}^T\mathbf{x} - (\mathbf{m}^T\mathbf{y} - f(\mathbf{y})).$$

Rearranging terms to compare $f(\mathbf{y})$ and $f(\mathbf{x})$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{m}^T(\mathbf{y} - \mathbf{x}).$$

This inequality holds for all $\mathbf{y} \in \mathbb{R}^n$. This is precisely the defining inequality for $\mathbf{m}$ being a subgradient of f at $\mathbf{x}$. Since f is differentiable, the subgradient is unique and equal to the gradient. Thus,

$$\mathbf{m} = \nabla f(\mathbf{x}).$$

□

Remark 4.2: *Geometrically, this corollary asserts that the vector* $\mathbf{m}$ *defines a supporting hyperplane to the graph of* f *at* $(\mathbf{x}, f(\mathbf{x}))$ *if and only if Fenchel's inequality holds with equality.*

4.1 Partial Minimization & Relation To Conjugate

Let $f : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R} \cup \{+\infty\}$ be a function of two variables $(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^k$.

Definition 4.3 (Partial Minimization): *The partial minimization of f with respect to $\mathbf{x}$ is the function $h : \mathbb{R}^k \to \mathbb{R} \cup \{+\infty\}$ defined as,*

$$h(\mathbf{y}) = \inf_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}, \mathbf{y}).$$

Lemma 4.4 (Preservation of Convexity): *If f is jointly convex in $(\mathbf{x}, \mathbf{y})$, then $h(\mathbf{y})$ is a convex function.*

Remark 4.5:

$$f^*(\mathbf{m}_1, \mathbf{m}_2) = \sup_{\mathbf{x},\mathbf{y}} \left(\mathbf{m}_1^T \mathbf{x} + \mathbf{m}_2^T \mathbf{y} - f(\mathbf{x}, \mathbf{y})\right)$$

Remark 4.6: $h^*(\mathbf{m}) = f^*(0, \mathbf{m})$

***Proof*:**

$$\begin{aligned}
h^*(\mathbf{m}) &= \sup_{\mathbf{y}} \left(\mathbf{m}^T \mathbf{y} - h(\mathbf{y})\right) \\
&= \sup_{\mathbf{y}} \left(\mathbf{m}^T \mathbf{y} - \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y})\right) \\
&= \sup_{\mathbf{y}} \sup_{\mathbf{x}} \left(\mathbf{m}^T \mathbf{y} - f(\mathbf{x}, \mathbf{y})\right) = f^*(0, \mathbf{m})
\end{aligned}$$

Remark 4.7: $\sup\left[-f^*(0, \mathbf{m})\right] \leq \inf f(\mathbf{x}, 0)$ *i.e.* $h^{**}(0) \leq h(0)$

4.1.1 Perturbation Interpretation Of Duality

For any problem $p^* = \min_{\mathbf{x}} f(\mathbf{x})$, "the" dual is constructed by

1. Introduce perturbations $\mathbf{y}$, and extend f such that $f(\mathbf{x}, 0)$ returns the function to minimize
2. the dual is $d^* = \sup\left[-f^*(0, \mathbf{m})\right]$

   where $\mathbf{m}$ is the dual variable and $\mathbf{x}$ is the primal variable.
   We have

$$d^* \leq p^*,$$

under convexity + mild conditions, we have strong duality, i.e. $d^* = p^*$.

5 Lagrangian Duality

Given the optimization problem in standard form:

$$p^* = \inf\{f_0(x) \mid f_i(x) \leq 0,\ i = 1, \ldots, m,\ x \in \mathcal{D}\}$$

where $\mathcal{D} \subseteq \mathbb{R}^n$ is the intersection of the domains of all functions f_i.

5.1 The Lagrangian and Dual Function

Define the **Lagrangian** $L : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ for $\lambda \succeq 0$ as,

$$L(x, \lambda) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x).$$

The **Lagrange dual function** $g : \mathbb{R}^m \to \mathbb{R}$ is defined as the pointwise infimum of the Lagrangian over x,

$$g(\lambda) = \inf_{x \in \mathcal{D}} L(x, \lambda).$$

By the property of pointwise infima, g is a concave function, and it provides a lower bound on the primal optimal value: $g(\lambda) \leq p^*$ for all $\lambda \succeq 0$.

The Dual Problem

The **Lagrange dual problem** seeks the tightest lower bound,

$$\begin{aligned} d^* = \sup_{\lambda} \quad & g(\lambda) \\ \text{s.t.} \quad & \lambda \succeq 0 \end{aligned}$$

The relationship between the primal and dual optimal values is governed by **Weak Duality** ($d^* \leq p^*$). Under constraint qualifications (e.g., Slater's condition) and convexity, **Strong Duality** holds ($d^* = p^*$).

6 Shadow Price Interpretation of Duality

7 Game Theoretic Interpretation of Duality

8 KKT Conditions

9 Mechanistic Interpretation of KKT Conditions